2026 MEISHA YOUTH HACKATHON INVITATIONAL · MES HACKATHON

Smart Campus Homework Analyzer

Turn every grading session into the next lesson

v0.2.0-rc1 · Team B3 · MIT License · Data stays on school premises

THE PROBLEM

The night grading finishes, the real work has just begun.

A teacher faces far more than just 'finishing the grading.'

Grading is time-consuming

Annotating each question, writing comments, recording scores. Whole blocks of time get fragmented — one class takes an entire evening.

Statistics are manual

Class error rates, knowledge mastery, error cause distribution — to see clearly, you have to count question by question.

Review relies on gut feeling

The next day in class, you can usually only address 'the ones that felt like most students got wrong.'

Result: **the part that exhausts teachers the most is precisely the part most easily wasted.**

THE INSIGHT

Grading produces data, but data never becomes a lesson.

The mistake notebook holds the class's most authentic learning evidence, yet no one has time to translate it into 'what to teach next class, who needs remediation.'

So what's truly missing isn't a faster grading tool — it's a translator that turns grading results into **teachable content**.

THE PRODUCT

Turn a stack of homework into a lesson.

Photo upload → AI page-by-page recognition & grading → Teacher review → Auto-generated class insights, knowledge mastery rates, error cause attribution, and ready-to-use review lesson plans.

Full-stack runnable | Local deployment | Data stays on campus | v0.2.0-rc1

Open it now, use it now.



HOW IT WORKS

Five-step loop, from a stack of paper to a lesson.

Every step has a human confirmation checkpoint.

01

Setup & Import

Independent school space; import up to 500 students at once via CSV / XLSX.

02

Create Assignment

Upload JPG / PNG / WebP / PDF / DOCX originals, parsed into editable questions.

03

Page-by-Page OCR

PDFs converted to page images server-side, page-by-page OCR, strictly aligned with student answer sheets.

04

Teacher Review

Low-confidence content requires teacher confirmation before entering student mistake notebooks.

05

Teaching Analysis

One-click review plans, error cause statistics, tiered student lists, and practice assignments.

Automation handles speed; **teachers handle judgment**.

AI TEACHING ANALYSIS · NEW IN V0.2.0

The moment review is done, the lesson plan is already there.

Not a longer report —
something you can bring into the classroom the next day.

Class Profile · Knowledge Mastery · Error Attribution · High-Freq Mistake Sheet · Lesson Plan · Tiered Support · Follow-Up Practice · One-Click Export



DESIGN PRINCIPLE

Numbers are always trustworthy; AI just puts them into plain language.

Deterministic Statistics · Verifiable

All computed from teacher-reviewed data.

- Only review-confirmed mistakes enter statistics; unreviewed ones counted separately
- Error causes strictly divided by question number; percentages always sum to 100%
- Questions marked 'Unrecognizable' listed separately, excluded from student error causes

The last rule prevents penalizing students for poor scan quality.

AI Transcription · Reviewable

Only transcribes and suggests; never generates facts.

- Core statistics remain fully functional when model is unavailable or API Key not configured
- Teachers can click 'AI Re-attribution' to re-evaluate with question context and knowledge points
- Generated review scripts, board notes, and diagnostics shown as-is, editable line by line

The cost of an error is 'a different phrasing,' not 'a wrong number.'

In short: **let the model do what it's good at — expression; let the code do what it's good at — arithmetic.**

KNOWLEDGE MASTERY & ERROR CAUSES

Which gap to fill first? The system tells you directly.

18 knowledge points sorted by accuracy rate low to high; error causes strictly classified by question number, 12 built-in rule categories, percentages always sum to 100%.



REVIEW SCRIPT

Not 'what was missed most' —
but 'how do I teach it.'

Sorted by number of students who missed it, starting with the most critical.

Expand to see affected students, reference answers, AI step-by-step solutions, and a review script you can read aloud.

Every high-frequency mistake includes

Explain to Students

One sentence that earns points

Board Demonstration

Content to write on the board

Quick Check After Explanation

Verify understanding in class





LESSON PLAN & DIFFERENTIATION

45 minutes — what to teach, who does what, all written out.

Theme, duration, objectives, timeline (what to teach / what the teacher does / what students do), and ready-to-copy board design.

Each error cause comes with a **5-minute actionable** classroom activity and a named list for tiered support.

Board Design · Error Intervention · Tiered List · One-Click Repractice

THE STUDENT SIDE

Every student gets their own copy.

Only teacher-confirmed mistakes enter the student notebook; knowledge points marked as passed once all answered correctly.



REAL DATA · MIDDLE SCHOOL PHYSICS · CHAPTER 15 STAGE ASSESSMENT

From a real exam paper, the system read what the teacher didn't have time to read.

87.5%
of mistakes were 'unanswered'
21 / 24 instances

18%
Class scoring rate
Average 18 / 100 pts

5
questions with 100% class error rate
Q1, 6, 15, 16, 17

36
reusable similar exercises
covering 18 knowledge points

AI Teaching Conclusion · Original Excerpt

"This Chapter 15 stage assessment had only 1 student (c) who submitted 2 times, totaling 3 records, 18 questions with 24 mistakes across 36 answer positions at 66.7% error rate, scoring only 18.0%. Most critically, 21 of the 24 mistakes were [Unanswered] (87.5%)... Points were lost mainly because students "didn't attempt" rather than "didn't know.""

ENGINEERING

Production-ready means actually usable.

Open-source MIT, locally deployable, data stays on campus.

Tech Stack

Frontend React 19 + TypeScript strict mode

Backend Express + SQLite (single-writer)

Documents PDF / DOCX / DOC page-by-page parsing, server-side image conversion

Models 10 provider presets, OpenAI-compatible vision format

Deployment Cloudflare Worker static hosting + API proxy

Verification

24 Vitest unit tests

Error classification · Statistical standards · Data isolation · AI attribution fusion

33 end-to-end verifications

Read/write APIs · Statistical standards · Permission boundaries; uses fixture data, no paid model calls

npm run build

Strict TS checks + production build

Security & Data

API Key encrypted server-side with AES-256-GCM

All queries isolated by **school and identity**

Upload type and size validation; originals and student data excluded from version control

Teacher review is a prerequisite for writing to mistake notebooks

Pre-launch TODO: email verification, login rate limiting, virus scanning, minor data processing terms.

OPEN SOURCE & STACK

Fully open source — tech choices laid bare for you to see.

MIT License, code on GitHub, one command to run it yourself.

Open Source · MIT

Source code, data formats, and statistical methods all public.



github.com/turtlelnc/MES-Homework

- **Self-deployable** — runs on a single school server, no hosted services required
- **Data stays on campus** — student originals, accounts, and database remain local
- **Extensible** — school knowledge points, error rules, and model providers all replaceable

Tech Choices · Minimal & Stable

No unnecessary dependencies for show — every layer independently replaceable.

Frontend

React 19 · TypeScript strict mode · Vite 6 · Recharts

Server & Data

Express 5 · SQLite (WAL) · Multer upload type and size validation

Documents & Recognition

PDF.js · mammoth · word-extractor · @napi-rs/canvas server-side image conversion · vision model OCR

Models & Security

10 provider presets · OpenAI-compatible vision format · API Key encrypted server-side with AES-256-GCM · scrypt password hashing

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Testing & Delivery · Vitest unit tests · node:test end-to-end verification · Cloudflare Worker static hosting & API proxy

WHAT'S NEXT

Let every grading session become the next lesson.

Next steps: admin approval for teacher onboarding, email verification and password recovery, login rate limiting, and recognition rate evaluation against real school samples.

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Team B3

2026 Meisha Youth Hackathon Invitational